

Hampus Serneke

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EDUCATION

École polytechnique fédérale de Lausanne, EPFL

M.Sc. in Data Science

- Incoming M.Sc. student at EPFL.

Lausanne, Switzerland

Sept. 2026 – July 2028

Lund University Faculty of Engineering

B.Sc. in Technology (as part of the Engineering Mathematics program)

- **Cumulative GPA:** 4.9/5.0
- Pursued a major in Computer Science and Engineering for two years before transferring to Engineering Mathematics.
- **Thesis title:** Deep Learning for Rotation Averaging in Structure from Motion
- **Coursework:** Mathematical statistics, matrix theory, stationary stochastic processes, applied mathematics (PDEs), algorithms and data structures, functional programming, object-oriented modelling, electromagnetic field theory

Lund, Sweden

Aug. 2022 – June 2026

Katedralskolan Lund

Natural Science High School program

- **GPA:** 22.5/22.5

Lund, Sweden

Aug. 2018 – June 2021

WORK & LEADERSHIP EXPERIENCE

California Institute of Technology, Caltech

Summer Undergraduate Research Fellow

- Conducting research in reinforcement learning and AI safety in robotics at the AMBER Lab under the supervision of Dr. Pol Mestres and Prof. Aaron Ames.

Pasadena, USA

June 2026 – Aug. 2026

Lund University

Teaching Assistant in Mathematics, Faculty of Engineering

- Led weekly exercise sessions where I taught calculus to a class of 30 engineering undergraduates.

Lund, Sweden

Aug. 2025 – Mar. 2026

Axis Communications

Machine Learning Intern, Core Technologies Analytics

- Worked to reduce bias in computer vision models using synthetic data.
- Developed a pipeline for generating synthetic data in Unreal Engine.
- Trained deep learning models for object detection and improved overall mAP, while reducing coverage bias by greatly improving performance on key object classes.
- Implemented a novel noise reduction algorithm for pre-processing synthetic data in Python.
- Streamlined the rendering pipeline in Unreal and refactored the codebase in C++.
- Collaborated across departments to generate data for different applications, onboarded consultants, and worked directly with a team of 10 engineers.
- Worked part-time during the terms and full-time during the summers.

Lund, Sweden

June 2023 – May 2025

Hemkodat

Coding Instructor

- Travelled in Sweden during holidays and weekends to deliver courses in Python, Lua, and C# for adolescents.
- Contributed to onboarding and training of new instructors.

Sweden

Sept. 2018 – Jan. 2023

The Computer Engineering Guild at LTH

Vice President

- Led the board of a student guild with 1,000 members, chaired board meetings, and managed administration.

Lund, Sweden

Jan. 2023 – Dec. 2023

SKILLS, PROJECTS & OTHER

Technical Skills: Python (PyTorch, TensorFlow), Java, C++, C#, Scala, Haskell, SQL, CI/CD, Git

Languages: English C2 (IELTS Academic 8.5/9.0), Swedish native

Personal Projects: Created a VR simulation of a student dorm in Unreal, built an automated candy dispenser with a Raspberry Pi, developed an algorithm for solving Ricochet robots in Java, built an NLP model with Keras, developed regression models for biological experiments

Open-Source Contribution: Implemented a more stable geodesic distance loss function in the PyRoMA framework for geometric deep learning with PyTorch

Additional Experience: Studied theoretical philosophy at Lund University and worked as a substitute teacher after graduating from high school

Awards: Won the national qualifiers and competed at the European Union Science Olympiad (Portugal 2019)